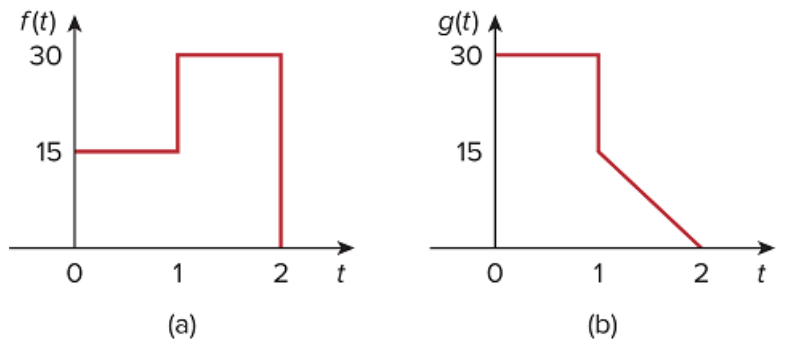


Problem

Find the Fourier transforms of both functions in Fig. 18.31 on the following page.

Figure 18.31



Step-by-step solution

Step 1 of 5

(a)

(Refer to Figure 18.31(a) in the textbook) The signal $f(t)$ is defined as,

$$f(t) = \begin{cases} 15, & 0 < t < 1 \\ 30, & 1 < t < 2 \end{cases}$$

Step 2 of 5

Take the Fourier transform of signal $f(t)$.

$$\begin{aligned} F(\omega) &= \int_0^1 15e^{-j\omega t} dt + \int_1^2 30e^{-j\omega t} dt \\ &= 15 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 + 30 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_1^2 \\ &= 15 \left[\frac{e^{-j\omega}}{-j\omega} + \frac{1}{j\omega} \right] + 30 \left[\frac{e^{-2j\omega}}{-j\omega} + \frac{e^{-j\omega}}{j\omega} \right] \\ &= \frac{15 + 15e^{-j\omega} - 30e^{-2j\omega}}{j\omega} \end{aligned}$$

Hence, the Fourier transform of signal $f(t)$ is $\boxed{\frac{15 + 15e^{-j\omega} - 30e^{-2j\omega}}{j\omega}}$.

Step 3 of 5

(b)

(Refer to Figure 18.31 (b) in the textbook) The signal $g(t)$ is defined as,

$$g(t) = \begin{cases} 30, & 0 < t < 1 \\ (30 - 15t), & 1 < t < 2 \end{cases}$$

Step 4 of 5

Take the Fourier transform of signal $g(t)$.

$$\begin{aligned} G(\omega) &= \int_0^1 30e^{-j\omega t} dt + \int_1^2 (30 - 15t)e^{-j\omega t} dt \\ &= 30 \int_0^1 e^{-j\omega t} dt + 30 \int_1^2 e^{-j\omega t} dt - 15 \int_1^2 te^{-j\omega t} dt \end{aligned}$$

The integration of xe^{ax} is,

$$\int xe^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Then, the signal $G(\omega)$ is,

$$\begin{aligned} G(\omega) &= \int_0^1 30e^{-j\omega t} dt + \int_1^2 (30 - 15t)e^{-j\omega t} dt \\ &= 30 \int_0^1 e^{-j\omega t} dt + 30 \int_1^2 e^{-j\omega t} dt - 15 \int_1^2 te^{-j\omega t} dt \\ &= 30 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 + 30 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_1^2 - 15 \left[\frac{e^{-j\omega t}}{(-j\omega)^2} (-j\omega t - 1) \right]_1^2 \\ &= 30 \left[\frac{e^{-j\omega}}{-j\omega} + \frac{1}{j\omega} \right] + 30 \left[\frac{e^{-2j\omega}}{-j\omega} + \frac{e^{-j\omega}}{j\omega} \right] - 15 \left[\frac{e^{-2j\omega}}{\omega^2} (2j\omega + 1) - \frac{e^{-j\omega}}{\omega^2} (2j\omega + 1) \right] \end{aligned}$$

Step 5 of 5

Rewrite the above equation.

$$\begin{aligned} G(\omega) &= \frac{-30}{j\omega} [e^{-j\omega} - 1] + \frac{30}{-j\omega} [e^{-2j\omega} - e^{-j\omega}] - 15 \left[\frac{e^{-2j\omega}}{\omega^2} (2j\omega + 1) - \frac{e^{-j\omega}}{\omega^2} (2j\omega + 1) \right] \\ &= \frac{-30j\omega + 30j\omega e^{-2j\omega} - 30j\omega e^{-j\omega} - 15e^{-2j\omega} + 15j\omega e^{-j\omega} + 15e^{-j\omega}}{\omega^2} \\ &= \frac{-30j\omega - 15e^{-2j\omega} + 15j\omega e^{-j\omega} + 15e^{-j\omega}}{\omega^2} \end{aligned}$$

Hence, the Fourier transform of signal $g(t)$ is $\boxed{\frac{-30j\omega - 15e^{-2j\omega} + 15j\omega e^{-j\omega} + 15e^{-j\omega}}{\omega^2}}$.